

Final Project: Conducting and Presenting Your Own Research

Physical Models: Anything you've learned in Physics I or II that you can come up with a way to investigate!

Analysis Tools: Anything you've learned this semester, or anything more advanced you've learned or chosen to learn from other classes or resources.

Experimental Systems: Anything from PHY 105N or systems you bring in yourself. **Equipment:** Anything from PHY 105N (also 105M by request)



Safety Concerns: You're encouraged to bring in items from outside of class for use in your investigation. But if you do so, please check with your TA to ensure your use of the item meets safety standards.

Introduction

Over the course of the semester, you've been introduced to a variety of measuring devices, quantitative analysis tools, physical models, and experimental systems. You've also been given some investigative freedom, allowing you to work out your own methods or choose your own objectives. Now it's time to take it to the next level and conduct an experiment of your own!

In this Final Project you'll design your own experiment and present it in the form of an (electronic) poster.

This is an individual assignment. You'll need to conduct your own investigation, collect your own data, and turn in your own poster. You are of course welcome to study the same topic as another student, and to discuss your approach with anyone, but your project must be done independently.

Part 1: Planning and Conducting Your Research

Deciding on a Project: The first thing you'll need to decide is what you'd like to investigate in your project. You're welcome to take as expansive a view on this as you like. We'll let you do anything as long as it's safe, substantive, and we can provide you with the necessary equipment. To help you decide, here are some options to consider:

- Was there anything you wanted to investigate during the semester but didn't get a chance to? For example, would you like to know if some additional household items (or locations on campus) are radioactive? If a car mirror has a well-defined focal point, and what it is? To what degree do different meter sticks differ from each other in length?
- In each lab we asked you to write ways to improve or build on an experiment in every conclusion, but not all of these were followed up on. Is there a lab where you'd be interested in doing a Part 3 based on your ideas for improvement?
- Was there a physical system we examined which you'd like to learn more about? A measuring device you'd like to play with more? An analysis method you'd like to take to the next level?

Focus on being creative. An important restriction is that whatever you do, it must genuinely differ from a lab we already did, including parts you chose not to do. (For example, if you chose not to examine the radioactive shielding in Lab 2, you **cannot** go back and do that).

Conducting Your Research: The second thing you'll need to do is conduct your research. During the last two weeks of class, we'll hold a special lab session. During this session, all of the lab setups from the whole semester will be available to you. You can also request additional equipment if we have it available, or equipment from Physics I lab. This is your opportunity to collect the data you need for your project.

Some projects may not require you to physically be in lab to complete the work. For example, you may be able to complete it using a phone app. If you find that you don't need to come in to lab to complete your project, we *highly recommend* that you discuss your project with your TA beforehand.

Part 2. Presenting Your Research

Poster Presentation: Like a regular lab, you'll need to document your method, data, and conclusion. But rather than presenting these as "Lab Notes", you'll integrate them into a Poster Presentation.

This Poster Presentation will also need to include an Abstract and Title. Refer to the writing assignment prelab and the feedback on your abstract from that week to help you think about how to compose this.

Your poster should include visual representations of data, like graphs with uncertainty bars, labels, equations, and other information, and should otherwise be visually well organized and appealing. See Prelab 1 for help with producing plots in excel, and the example posters for ideas on good presentation.

Turning it in: You'll turn in an electronic copy of your poster presentation to your TA via Canvas. It'll be due on the last day of classes of the semester (see the prelab/registrar for this date).

Grading Criteria

For Methods, Data, and Conclusion you'll be graded by the same criteria as a (single part of) Lab Notes assignment, except each category will be worth 10 points.

For the Abstract and Title, you'll be graded by the same criteria as the writing assignment.

Additionally, you'll be given a score out of 10 for both presentation and creativity. So the rubric for the project will look like this:

Abstract (0-10)

Presentation and Visualizations (0-10)

Methods (0-10)

Data (0-10)

Conclusion (0-10)

Creativity and Originality (0-10)

This totals to 60 points. It will be worth 20% of your final grade